SPiD
Sustainable Participatory Development and Climate Resilience Initiative

VOICES FROM THE WARD

Climate Vulnerability, Awareness, and Governance in Sabon Gari LGA, Kaduna State

A Quantitative and Geospatial Assessment



Community-led Participatory Initiative for Climate Governance (CPI4CG)

Sustainable Participatory Development and Climate Resilience Initiative (SPiD)

NESTAFRICA AI Innovation Lab

With technical support on GIS, Data Management, and Analysis from Nest Africa Artificial Intelligence Innovation Lab

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PREFACE

Background of the Organization and Project

About SPiD

The **Sustainable Participatory Development and Climate Resilience Initiative (SPiD)** is a youth-led, community-driven non-governmental organization established in 2024 and registered in Nigeria. It works at the intersection of climate resilience, sustainable development, environmental sustainability, inclusive governance, and community empowerment, advancing locally driven, evidence-based approaches that recognize communities, particularly young people, women, and underserved populations, as essential actors. Through research, capacity building, policy advocacy, stakeholder engagement, and strategic partnerships, SPiD works to strengthen participatory climate governance and locally led climate resilience in Nigeria.

About CPI4CG

The **Community-led Participatory Initiative for Climate Governance (CPI4CG)** is an initiative of SPiD that strengthens community participation in climate governance through evidence generation, inclusive stakeholder engagement, capacity building, and evidence-based advocacy. Nigeria has established frameworks such as the Climate Change Act (2021), the National Adaptation Plan Framework, and its Nationally Determined Contributions (NDCs), yet significant challenges remain in translating these commitments into locally responsive climate action. Communities live with changing rainfall, extreme temperatures, flooding, drought, and declining agricultural productivity, but their experiences, priorities, and adaptation practices are not always reflected in climate planning, resource allocation, and decision-making.

CPI4CG is piloted in **Sabon Gari Local Government Area, Kaduna State**, with the broader ambition of developing a replicable model for community-led climate governance across Nigeria.

The Importance of Participatory Research in CPI4CG

Participatory research is the foundation of CPI4CG. It treats community members as contributors to knowledge generation, problem identification, solution development, and decision-making, rather than merely as respondents. The research was designed around household surveys, Focus Group Discussions (FGDs), and Key Informant Interviews (KIIs) covering five interconnected areas:

1. Climate change awareness and perception
2. Climate impacts and local experiences
3. Climate governance and institutional gaps
4. Traditional and indigenous ecological knowledge
5. Pathways for strengthening community participation

Its findings are intended to inform community validation and feedback sessions, stakeholder dialogues, policy recommendations, and advocacy, so that communities help shape solutions and influence the institutions responsible for addressing them.

Acknowledgements

SPiD sincerely thanks the **Sabon Gari Local Government Authority** and the departments and officials who facilitated engagement with communities and local institutions. We are grateful to the **District Heads, Traditional Leaders, and Community Leaders** across the 11 wards, whose guidance on community entry built trust and local ownership.

Special appreciation goes to **Nest Africa AI Innovation Lab**, particularly **Abubakar Isa Abubakar** and **Bashir Abdulraheem**, for technical collaboration on the data collection tools and the data analysis. We recognize the **volunteer data collectors** who, after capacity-building training, engaged community members across the 11 wards.

Our deepest appreciation goes to the **368 community members** who shared their experiences, knowledge, concerns, and recommendations. Behind every response is an individual, household, or livelihood whose experience deserves to be heard in climate decision-making.

We also thank our research adviser, **Ibrahim Muhammad Shamsuddin**, and SPiD's project team, particularly the Project Lead, **Abdulbasit Salahudeen Mikail**, Founder and CEO of SPiD, and the wider project implementation team.

This research represents a collective effort.

SUMMARY

Executive Summary

Between November and December 2025, the Sustainable Participatory Development and Climate Resilience Initiative (SPiD), in partnership with the Sabon Gari Local Government Area in Kaduna State, the District Heads of Sabon Gari LGA, and the Nest Africa Artificial Intelligence Innovation Lab, under its Community-led Participatory Initiative for Climate Governance (CPI4CG), surveyed 365 households across all eleven wards of Sabon Gari Local Government Area (LGA), Kaduna State, and complemented the survey with a geospatial assessment of land degradation using satellite-derived Trends.Earth data. The study conducted quantitative survey and GIS data collection; every finding in this report is grounded strictly in those two evidence streams.

The evidence describes a community that is exposed, aware, and willing but under-resourced, under-informed about institutional pathways, and largely excluded from formal climate governance. Households reported losing an average of 36% of income to climate hazards in the last 12 months, with flooding the single most-cited main hazard (28% of respondents) and flood-related hazards combined (flood, river overflow, erosion) named by 41%. The geospatial evidence corroborates this vulnerability: 39.4% of the assessed land area is degraded under SDG indicator 15.3.1, and artificial surfaces more than doubled between the baseline (2003–2015) and reporting (2015–2022) periods: a +254 km² expansion drawn entirely from cropland, consistent with the drainage congestion and urban flood exposure respondents describe.

Awareness of climate change is high (80%), but capability is thin: only 12% report high understanding of causes or of household risk-reduction actions, 41% have taken any adaptation action, 19% have attended a climate meeting in the past year, and just 20% know which government office to contact for climate support. Among households that acted, lack of money was the dominant barrier (57%); among those that did not, 100% said they would adopt new practices if supported. Governance perceptions split sharply: residents trust proximate institutions (65% agree local leaders listen; 89% of those who report problems go first to a traditional leader) but distrust formal response (only 33% agree the LGA responds promptly; only 20% agree climate resource allocation is transparent). Willingness to engage is near-universal: 99% would join climate initiatives in the next six months, and 96% made a concrete household pledge.

The report recommends a ward-anchored participatory climate governance architecture: standing ward climate committees linked to traditional leadership; radio-first early warning and advisory services; targeted flood control and drainage investment in the highest-loss wards (Chikaji, Zabbi, Bomo, Muchia); micro-financing and savings-group strengthening to convert willingness into action; gender-intentional participation mechanisms; and integration of the SDG 15.3.1 evidence into LGA and state land-use and restoration planning under Nigeria's Climate Change Act (2021), National Adaptation Plan framework, and the AU Climate Change and Resilient Development Strategy (2022–2032).

365
households surveyed across all eleven wards

36%
average income lost to climate hazards in the last 12 months

39.4%
of the assessed land area is degraded under SDG indicator 15.3.1

20%
know which government office to contact for climate support

99%
would join climate initiatives in the next six months

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Acronyms and Abbreviations

ACRONYM	MEANING
AU	African Union
CPI4CG	Community-led Participatory Initiative for Climate Governance
CSO	Civil Society Organisation
EWS	Early Warning System
GIS	Geographic Information System
GPS	Global Positioning System
IPCC	Intergovernmental Panel on Climate Change
LDN	Land Degradation Neutrality
LGA	Local Government Area
LPG	Liquefied Petroleum Gas
NAP	National Adaptation Plan
NDC	Nationally Determined Contribution
NiMet	Nigerian Meteorological Agency
PWD	Person(s) with Disability
SAR	Synthetic Aperture Radar
SDG	Sustainable Development Goal
SEMA	State Emergency Management Agency
SPiD	Sustainable Participatory Development and Climate Resilience Initiative
TEK	Traditional Ecological Knowledge
UNCCD	United Nations Convention to Combat Desertification
WASH	Water, Sanitation and Hygiene

INTRODUCTION

1. Introduction

Climate change is among the defining development challenges of the 21st century, with disproportionate consequences for communities in the Global South whose adaptive capacity is constrained by poverty, infrastructure deficits, and weak institutional support (IPCC, 2022). Nigeria faces intersecting hazards: flooding, drought, desertification, heat stress, and pest pressure, whose cascading effects undermine livelihoods, food security, and public health. National frameworks such as the Climate Change Act (2021), the National Adaptation Plan (NAP) Framework (2020), and the revised Nationally Determined Contributions (NDCs) all invoke participation, gender equity, and locally led action, and regional instruments including the African Union's Agenda 2063 and the AU Climate Change and Resilient Development Strategy (2022–2032) emphasise community-level preparedness. Yet an implementation gap persists between these high-level commitments and locally relevant action, particularly at the Local Government Area (LGA) and ward levels where climate impacts are experienced.

This report presents the quantitative and geospatial evidence generated by the research component of the Community-led Participatory Initiative for Climate Governance (CPI4CG) in Sabon Gari LGA, Kaduna State. It documents household-level climate awareness, hazard exposure and impacts, adaptive practices, governance perceptions, and willingness to participate, and it situates these findings against satellite-derived evidence on land degradation and land cover change. The purpose is to give government agencies, development partners, civil society, and communities themselves a rigorous, shared evidence base for participatory climate governance.

> a rigorous, shared evidence base for participatory climate governance

2. Project Background

The Community-led Participatory Initiative for Climate Governance (CPI4CG) is implemented by the Sustainable Participatory Development and Climate Resilience Initiative (SPiD) in partnership with Nest Africa Artificial Intelligence Innovation Lab to strengthen community participation in climate governance through evidence generation, stakeholder engagement, and policy advocacy. The initiative responds to a well-documented pattern in Nigerian climate governance: communities are frequently treated as passive beneficiaries rather than co-producers of adaptation solutions, and provisions for participation in instruments such as the Climate Change Act (2021) remain weakly operationalised at LGA level.

The research component was designed as a mixed-methods participatory study combining a household survey, focus group discussions, key informant interviews, and geospatial analysis. Owing to implementation realities, only the quantitative household survey and the GIS component were completed. All analysis, interpretation, and recommendations in this report are therefore grounded strictly in the quantitative and geospatial datasets; no qualitative findings are inferred. The Data Management and GIS analysis was conducted in collaboration with the Nest Africa AI Innovation Lab using Trends.Earth land degradation data and satellite time-series imagery of the Shika Dam.

INTRODUCTION

3. Research Objectives

The general objective of the study is to strengthen community participation in climate governance by assessing awareness levels, climate impacts, governance gaps, and adaptation practices in Sabon Gari LGA, and to identify pathways for integrating community voices into formal climate governance processes. Within the scope supported by the available data, the specific objectives are to:

01 Assess the level of climate change awareness and self-rated understanding of causes and household risk-reduction actions among community members;

02 Document the socioeconomic profile and multidimensional vulnerability of households, including income, livelihoods, WASH, energy, and the situation of women and persons with disabilities;

03 Quantify exposure to climate hazards, their reported impacts, and self-estimated income losses across the eleven wards;

04 Examine access to early warning information, preferred communication channels, and institutional reporting pathways;

05 Measure adaptation practice, its perceived effectiveness, and the barriers that constrain further action;

06 Assess perceptions of climate governance, voice, responsiveness, transparency, and reliability, and willingness to participate;

07 Analyse land degradation and land cover change using SDG indicator 15.3.1 outputs, and integrate the geospatial evidence with survey findings;

08 Generate policy-relevant recommendations for community, local, state, national, civil society, and development partner actors.

METHODOLOGY

4. Methodology

4.1 Study Design

The study employed a cross-sectional quantitative household survey complemented by a geospatial (GIS) assessment. The original mixed-methods design included qualitative components (focus group discussions and key informant interviews) which were not implemented; the study is therefore reported and interpreted as a quantitative and geospatial assessment. The survey instrument was structured, administered face-to-face by trained enumerators using digital data collection with GPS capture, and covered household demographics, WASH and energy, climate awareness, hazard exposure and impacts, adaptation, governance perceptions, agriculture, health, early warning, and willingness to participate.

4.2 Study Area

Sabon Gari LGA lies within the Zaria metropolitan area of Kaduna State, Northern Nigeria, in the Northern Guinea Savanna zone. It is a peri-urban LGA combining dense settlement and market activity (Samaru, Sabon Gari town) with active cropland and wetland (fadama) areas along drainage lines and around the Shika Dam. Surveyed households span latitudes 11.10–11.22°N and longitudes 7.61–7.77°E. All eleven wards were covered: Bassawa, Bomo, Basawa-adjacent Bomo communities, Chikaji, Dogarawa, Hanwa, Jama'a, Jushi, Muchia, Samaru, Unguwan Gabas (Angwa Gabas), and Zabbi.

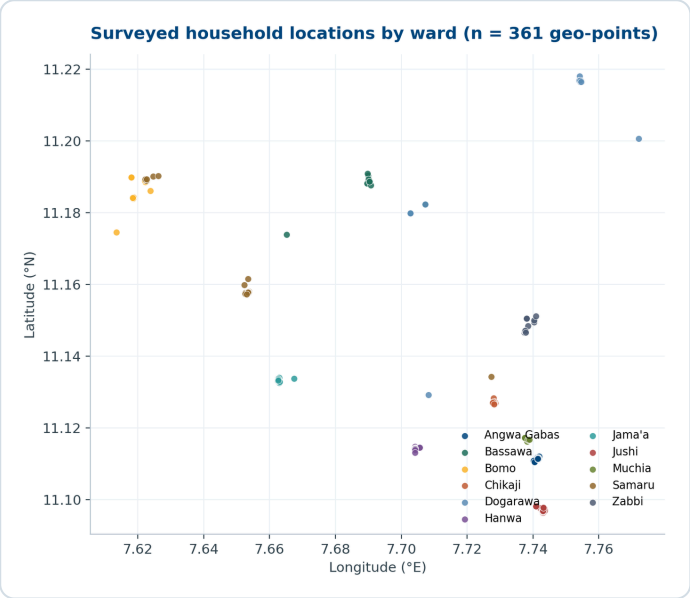


Figure 1. Spatial distribution of surveyed households by ward (GPS points captured at interview, n = 361).

365
valid household interviews completed

11
wards covered, with 29–37 interviews per ward

361
households with a GPS geopoint captured at interview

±5%
approximate margin of error at the 95% confidence level

4.3 Sampling

A multi-stage cluster approach guided sampling: wards within the LGA, communities within wards, and households within communities. A total of 365 valid household interviews were completed, distributed nearly evenly across the eleven wards (29–37 interviews per ward), which supports LGA-wide description and ward-level comparison. Respondents were adult household members; 51% were female and 49% male. For a population the size of Sabon Gari LGA, a sample of 365 corresponds to approximately a ±5% margin of error at the 95% confidence level under simple random sampling assumptions; design effects from clustering may widen this margin, so ward-level estimates should be read as indicative.

4.4 Data Collection

Fieldwork took place over approximately two weeks in November–December 2025. Enumerators administered the structured questionnaire in person, primarily in Hausa, recording responses digitally. Informed consent was obtained and recorded for every interview (100% consent; 99% also consented to recontact for validation). A GPS geopoint was captured at 361 of 365 households, enabling spatial integration with the GIS analysis. A visual aid (image card) was used to standardise the explanation of climate change for respondents who had not heard the term.

4.5 Quantitative Methods

Analysis used descriptive statistics (frequencies, percentages, means, medians) for all closed-ended items, with multiple-response items (hazards observed, impacts, barriers to participation, coping strategies, information sources, assets, inputs) analysed as the share of respondents selecting each option. Ranking items (at-risk groups; ward priorities) were analysed by first-ranked choice. Cross-tabulations disaggregated key indicators by gender, ward, and income band. Percentages are reported against the full sample (n = 365) unless a different valid base is stated. Self-estimated income loss was analysed as a continuous percentage.

Table 1. *Completed interviews by ward*

Ward ↕	Interviews ↕	Share of sample ↕
Bassawa	37	10.1%
Jushi	36	9.9%
Jama'a	34	9.3%
Chikaji	34	9.3%
Hanwa	34	9.3%
Angwa Gabas (Unguwan Gabas)	34	9.3%
Muchia	34	9.3%
Samaru	33	9.0%
Bomo	30	8.2%
Dogarawa	30	8.2%
Zabbi	29	7.9%
Total	**365**	**100%**



Photo. Training of volunteer data collectors: what data collection involves

4.6 GIS Methods

The geospatial component, conducted with Nest Africa AI Innovation Lab, used the Trends.Earth platform to compute SDG indicator 15.3.1 (proportion of land that is degraded over total land area) for the Sabon Gari analysis extent, comparing a baseline period (2003–2015) with a reporting period (2015–2022). Sub-indicators covered land productivity dynamics and land cover transitions across seven classes (tree-covered, grassland, cropland, wetland, artificial, bare land, water body). Seasonal and long-term surface water dynamics at the Shika Dam were examined using a satellite time-lapse combining Sentinel-1 SAR and Landsat imagery (from 2015) with Google Earth reference imagery of the dam, spillways, and downstream reaches.

4.7 Data Analysis

Survey data were cleaned (whitespace and category harmonisation; numeric coercion for household size, income loss, farm size, water insecurity days) and analysed in Python (pandas). GIS outputs were taken from the Trends.Earth summary tables and the accompanying land degradation mapping of the nine mapped ward areas (Basawa, Dogarawa, Zabi, Samaru, Muchiya, Ungwan Gabas, Jushi, Chikaji, Hanwa). Survey and GIS evidence were integrated analytically by juxtaposing ward-level survey patterns (hazard type, income loss) with the spatial degradation and land cover change evidence.

4.8 Ethical Considerations

Written or verbal informed consent was obtained from all respondents prior to the interview, and separate permission was sought for recontact during validation. Participation was voluntary, and community entry followed engagement with local structures. Respondent names and GPS coordinates are held for validation purposes only; this report presents aggregated, de-identified results, and the household location figure is rendered at a scale that does not identify individual dwellings.

4.9 Study Limitations

Absence of the qualitative component.

Planned focus groups and key informant interviews were not conducted; findings on motivations, indigenous knowledge content, and institutional dynamics are therefore limited to what closed-ended items can capture, and no qualitative claims are made.

Self-report and recall bias.

Income loss, hazard frequency, and illness measures are self-estimated over a 12-month recall window and should be read as perceived magnitudes rather than audited losses.

Cross-sectional design.

The survey is a single snapshot (Nov–Dec 2025, early dry season); seasonality may influence hazard salience — for example, recall of the just-ended rainy season may elevate flood reporting.

Ambiguity in one trend item.

Responses to "Have hazards become more frequent in the last 5 years?" used mixed response labels (Yes / Less / Same / Inconsistent), which limits interpretation of perceived trend direction.

GIS boundary extent.

The Trends.Earth assessed area (5,679 km^2) is substantially larger than the commonly cited administrative area of Sabon Gari LGA, indicating the analysis boundary covers a wider Zaria-region extent. Proportions (e.g., 39.4% degraded) describe that assessed extent and should be re-run on a verified LGA boundary before being cited as official LGA statistics.

Small data-quality artefacts.

A small number of records contain implausible or placeholder values (e.g., household size of 0, placeholder community names); these do not materially affect aggregate results but are noted for transparency. Ward-level means are based on 29–37 observations and carry wide uncertainty.

METHODOLOGY

From the Field

Photographs from CPI4CG fieldwork in Sabon Gari LGA. Select any photo to view it full size and browse the full set.


A conversation with women outside a compound


A mud-brick house with a collapsed wall and roof


An eroded drainage gully beside a hand-dug well


Young men at a hand-pump borehole


A drainage channel blocked with solid waste


Training of volunteer data collectors: downloading survey forms from the server


Men gathered for a community session


Women gathered for a community session in a mud-walled room


Mud-brick homes with peppers drying on the rooftops



Photo · Women gathered under a veranda, some holding paper forms

SECTIONS 5–6

Findings

5. Respondent Profile

6. Results

6.1 Climate Awareness, Knowledge, and Information

6.2 Hazard Exposure and Perceived Severity

6.3 Impacts on Livelihoods, Food Security, and Health

6.4 Adaptation Practice and Barriers

6.5 Governance, Voice, and Participation

6.6 Willingness, Priorities, and Pledges

FINDINGS

5. Respondent Profile

The sample is gender-balanced (51% female, 49% male) and concentrated in the productive age range: 53% of respondents are aged 20–40 and a further 30% aged 41–60, with 10% above 60. Most respondents are married (85%); 6% are widowed, nearly all of them women. Households are large; the mean household size is 9.5 members (median 9), meaning the 365 surveyed households collectively represent roughly 3,500 people.

Livelihoods are strongly agrarian for a peri-urban LGA: 58% of respondents identify as farmers, and crop farming is the primary household livelihood for 56% of households, followed by petty trade (18%). Farm sizes are small (mean 0.96 ha among cultivating households), 59% of cultivators lack irrigation access, and only 35% of all households received extension advice in the past year. Educational attainment is modest and gendered: 22% of respondents have no formal education (29% of women versus 15% of men), while 34% completed senior secondary and 14% tertiary education.

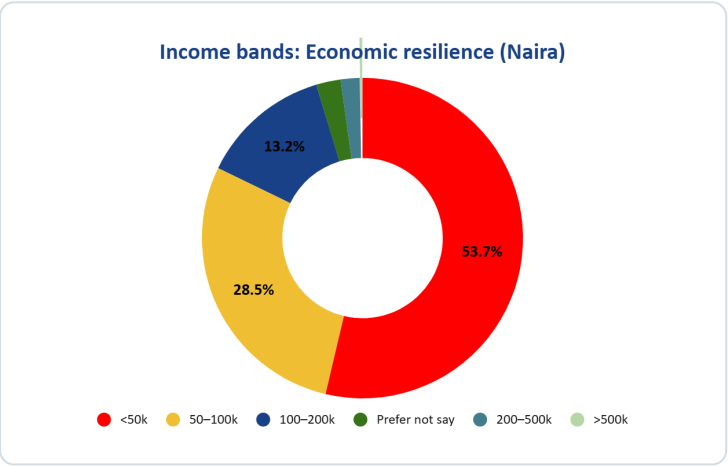


Figure 2. Distribution of average monthly household income bands (n = 365).

Economic precarity is the defining feature of the profile. Fifty-four percent of households report average monthly income below ₦50,000, and 82% fall at or below ₦100,000, for households averaging nearly ten members. Only 22% of households have any financial buffer against shocks, overwhelmingly informal savings groups (adashe-type) rather than insurance. Vulnerable-group presence is significant: 20% of households include at least one person with disability, 9% are female-headed, and 7% of respondents themselves live with a disability.

Table 2. *Household services and living conditions*

Indicator	Finding
Drinking water (dry season)	Borehole 55%; protected well 32%; piped 4%; unprotected well 4%
Drinking water (rainy season)	Borehole 39%; rainwater 30%; protected well 24%; unprotected well 3%
Days per month without enough safe water	Mean 3.0 days (range 0–40)
Sanitation	Pit latrine 83%; flush 12%; VIP latrine 2%; open defecation 3%
Cooking energy	Firewood 50%; charcoal 47% (biomass combined 97%); LPG 3%
Electricity	Grid (NEPA) 87%; none 9%; solar 2%; generator 1%
Key assets owned	Phone 80%; radio 48%; TV 47%; smartphone 42%; motorcycle 28%

Why this matters

Near-total dependence on biomass cooking (97%) ties household welfare directly to tree cover and land condition — the same land the GIS analysis shows is degrading — while the 30% who switch to rainwater for drinking in the rainy season face heightened water-quality risk during the flood season, consistent with the 72% of households reporting diarrhoeal or heat-related illness in the past year.

22%
of households have any financial buffer against shocks

20%
of households include at least one person with disability

9%
of households are female-headed

9.5
mean household size (median 9)

FINDINGS

6. Results

6.1 Climate Awareness, Knowledge, and Information

Awareness of climate change is high but shallow. Eighty percent of respondents have heard of climate change — and women report higher awareness than men (85% versus 75%) — yet self-rated understanding collapses at the next step: only 12% rate their understanding of the causes of climate change as high or very high, and only 12% rate their understanding of household risk-reduction actions as high or very high, while 43–44% rate each as low or very low. Fifty-eight percent attribute their main local hazard to climate change; 14% do not know the driver, and 11% cite "other" drivers, including religious attributions and locally specific causes such as blocked drainage and waste dumping recorded verbatim by enumerators.

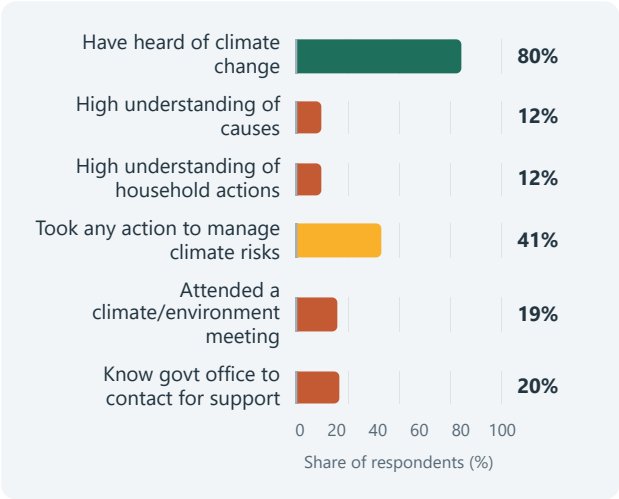


Figure 3. The awareness-to-action gap: share of respondents at each stage from awareness to institutional connection (n = 365).

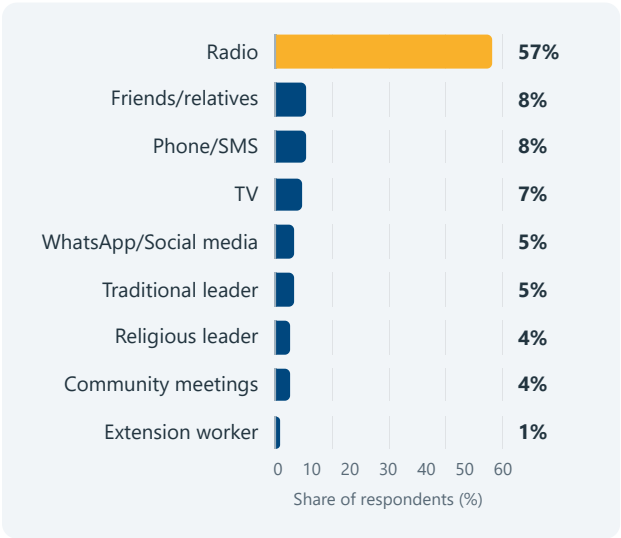


Figure 4. Preferred channels for climate alerts and advice (n = 364).

Information access is comparatively strong and overwhelmingly radio-borne. Seventy percent report access to timely weather or early warning information, and 65% received at least one early warning in the past year; of those warned, 85% cite radio as the source and 43% received warnings with more than seven days of lead time. Radio is also the preferred future channel for alerts and advice (57%), far ahead of interpersonal channels, phone/SMS, television, and social media. Extension workers are almost absent from the information ecosystem (1% preferred; 1% actual source).

Notable finding

Only 20% of respondents know which government office to contact for climate-related support, the single weakest link in the awareness chain. The community's information system relies on radio and traditional leaders; the state's formal support system is effectively invisible at the household level.

6.2 Hazard Exposure and Perceived Severity

Hazard exposure is broad-based and multi-hazard. Asked which hazards they had noticed in the last 12 months (multiple responses), respondents most frequently cited pest and disease outbreaks (47%), flooding (43%), windstorms (42%), and heatwaves (41%), with drought (27%), erosion (21%), and river overflow (19%) also widespread. When asked to name the single main hazard, flooding leads (28%), followed by pest/disease outbreaks (24%) and windstorms (15%); taking flood-related hazards together (flood, river overflow, erosion), 41% of respondents identify a flood-family hazard as their principal threat. Severity perceptions are substantial: 45% describe the main hazard as strong or extreme.

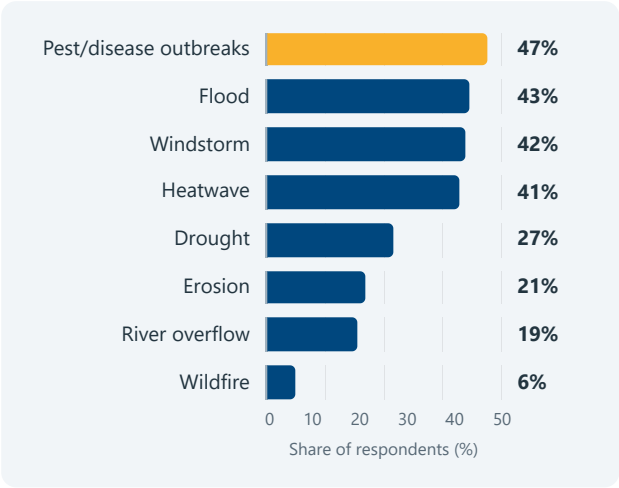


Figure 5. Hazards observed in the last 12 months, multiple responses (n = 365).

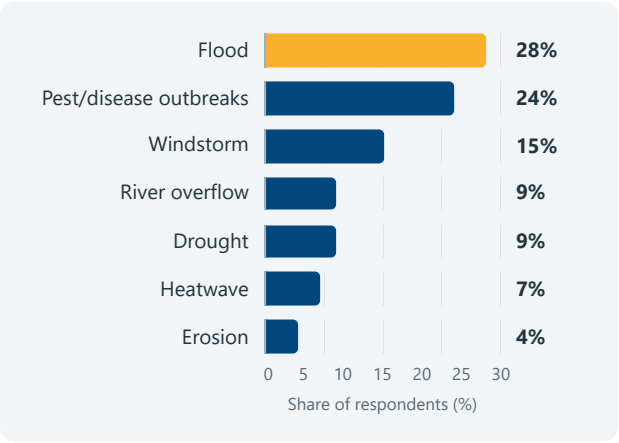


Figure 6. Main hazard of the last 12 months as identified by respondents (n = 351 naming a hazard).

Hazard profiles differ meaningfully by ward, consistent with the LGA's mixed peri-urban geography. Flooding is the modal main hazard in six wards (Angwa Gabas, Chikaji, Hanwa, Jama'a, Jushi, Muchia) — largely the denser, wetland- and drainage-adjacent areas — while pest/disease outbreaks dominate in the more agricultural wards of Bomo, Dogarawa, Samaru, and Zabbi, and windstorm leads in Bassawa. Perceptions of hazard-frequency trends are mixed and, given the response-label ambiguity noted in the limitations, no directional trend claim is made from this item.

41%
identify a flood-family hazard as their principal threat

28%
name flooding as the single main hazard

45%
describe the main hazard as strong or extreme

6.3 Impacts on Livelihoods, Food Security, and Health

The impacts of the main hazard are pervasive and material. House damage is the most commonly reported impact (59% of respondents), followed by injury or illness (50%), crop loss (49%), income loss (36%), and food shortage (33%). Nearly one in five households lost livestock (19%), 15% experienced school disruption, and 10% reported displacement. Health burdens extend beyond hazard events: 72% of households experienced diarrhoeal or heat-related illness in the past year, and 25% report that extreme heat affected work or school attendance.

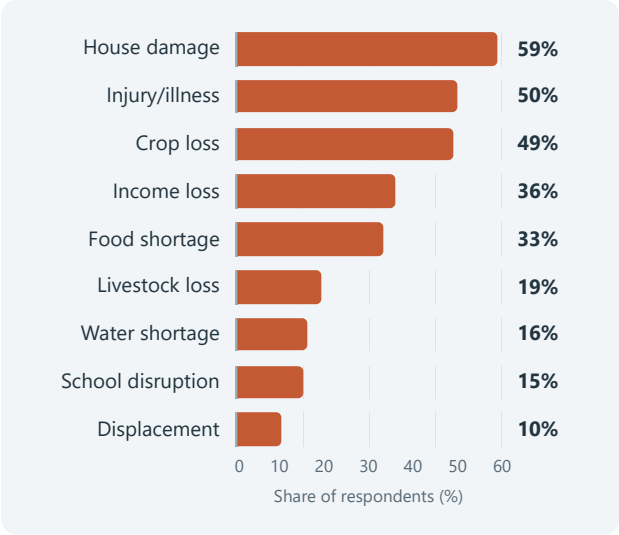


Figure 7. Reported impacts of the main hazard, multiple responses (n = 365).

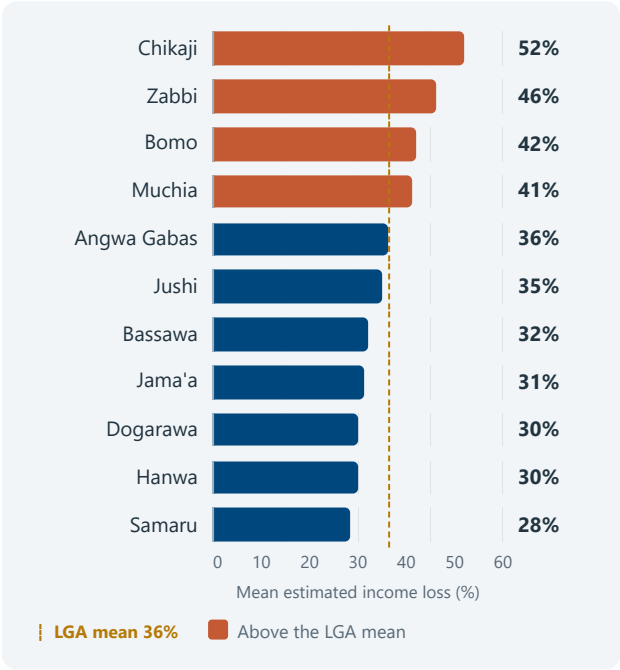


Figure 8. Mean self-estimated income loss from hazards by ward (n = 365; ward n = 29–37).

Households estimate losing, on average, 36% of income to hazards over the last 12 months (median 30%). Losses vary widely by ward, from a mean of 28% in Samaru to 52% in Chikaji, with Zabbi (46%), Bomo (42%), and Muchia (41%) also above the LGA mean. Losses also vary by hazard type: households naming erosion (mean 45%), pest/disease (40%), or drought (40%) as the main hazard report the deepest losses, while heatwave-affected households report the least (20%). Men report somewhat higher mean losses than women (40% versus 33%), plausibly reflecting men's greater concentration in farming as a stated occupation.

Notable finding

An average income loss of 36% in a population where 54% of households already live on under ₦50,000 per month implies climate hazards consume a large share of subsistence-level income, with only 22% of households holding any savings-group or insurance buffer to absorb it.

Agricultural results reinforce the pressure on the food system. Among cultivating households, 36% report yields lower or much lower than their five-year average, against 25% reporting higher; 29% could not judge. Input use is intensive (99% of input-reporting farmers used fertiliser, 91% organic manure, 85% pesticides), and heavy pesticide use is consistent with pest/disease outbreaks being the most widely observed hazard. Soil conservation is practised but partial: among the 239 households reporting a technique, mulching (51%), terracing (35%), cover crops (28%), and contour bunds (23%) lead.

6.4 Adaptation Practice and Barriers

Action lags awareness. Forty-one percent of households (148 of 365) took any action to manage climate risks in the past year. The most common actions were waste management and clean-ups (37% of acting households), tree planting and agroforestry (23%), changed planting dates (22%), mulching and soil conservation (20%), and water harvesting (20%). Households that acted judge their measures only moderately effective (53% "moderate"; 16% high or very high).

The binding constraint is money, not motivation. Among acting households, 57% name lack of money as the main barrier to doing more, far ahead of lack of support from authorities (18%) and lack of information (14%). Most strikingly, every one of the 217 households that took no action said they would adopt new practices if supported — a 100% latent-demand signal for adaptation support.

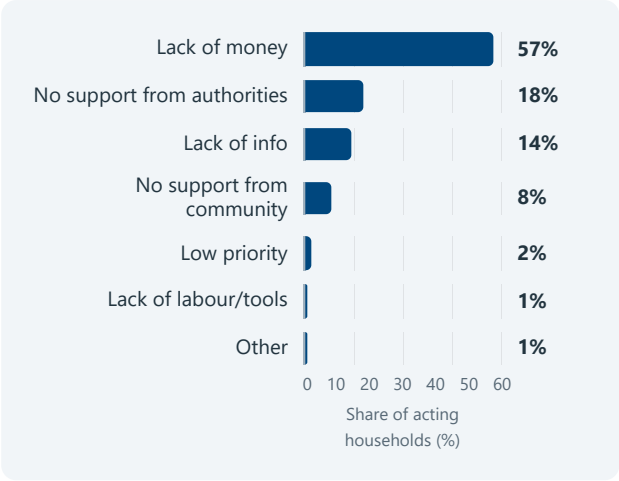


Figure 9. Main barrier to acting more, among households that took adaptation action (n = 148).

6.5 Governance, Voice, and Participation

Perceptions of climate governance divide cleanly between proximate and formal institutions. Majorities agree that local leaders listen to community concerns (65%), that women and youth have a voice in local climate decisions (68%), and that information from local authorities is reliable (65%). Confidence collapses where formal responsiveness and public finance are concerned: only 33% agree the local government/LGA responds promptly to climate impacts (47% disagree), and only 20% agree that climate resource allocation is transparent (57% disagree — the worst-rated item).

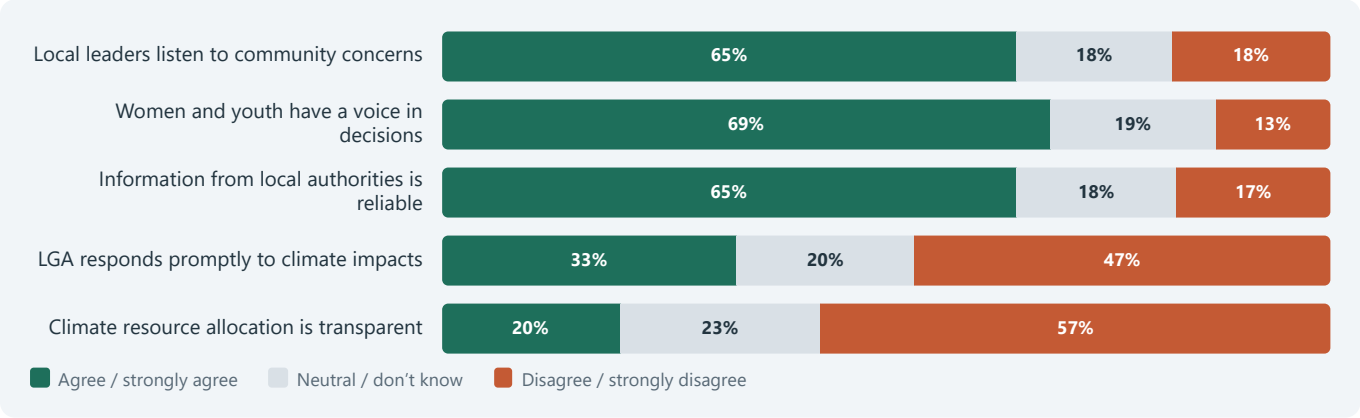


Figure 10. Perceptions of climate governance: agreement with five statements (n = 365).

Institutional participation is thin. Only 34% say a local group or committee responsible for environment/climate reporting exists (55% say none; 12% do not know), and only 19% attended any climate or environment meeting in the past year (attendees averaged about five meetings). Reporting pathways run through customary authority: of the 298 respondents who report climate-related problems somewhere, 89% go first to a traditional leader, 20% to a religious leader, and only 14% to the LGA office. The most cited barriers to participation for marginalised groups are lack of information (79% of respondents), lack of invitation (47%), time and care burdens (22%), cultural norms (21%), and language/literacy (16%).

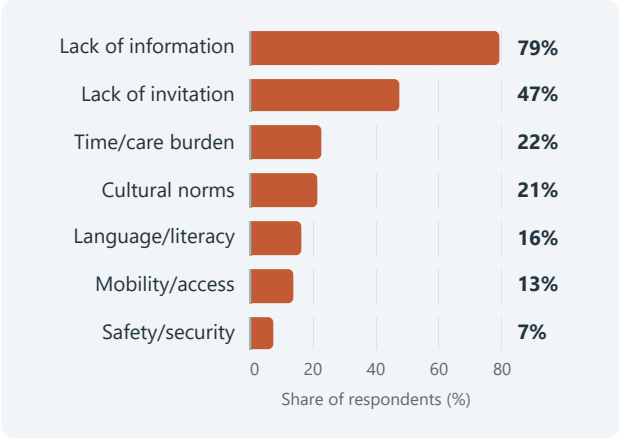


Figure 11. Barriers to participation in climate decision-making, multiple responses (n = 365).

Gender patterns cut in both directions. Women report higher climate awareness (85% versus 75%) and are markedly more likely to have taken adaptation action (49% versus 32%) — yet they are less likely to have attended a climate meeting (14% versus 24%) and nearly twice as likely to have no formal education (29% versus 15%). Women are acting more and deciding less.

> **Women are acting more and deciding less.**

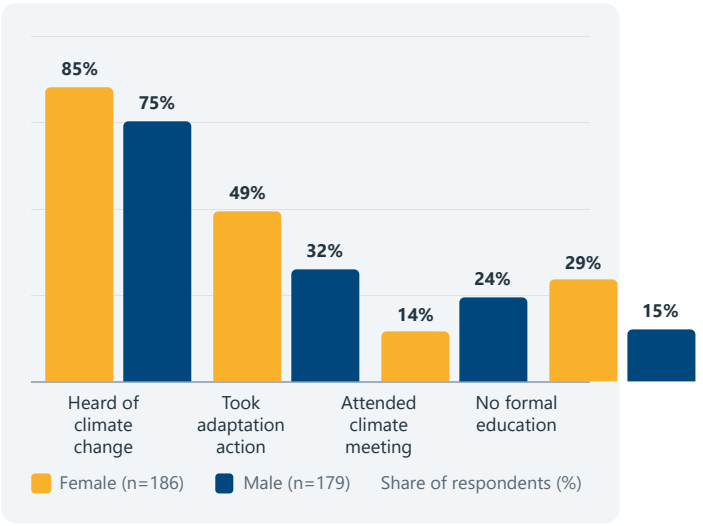


Figure 12. Gender differences in awareness, action, meeting attendance, and education (female n = 186; male n = 179).

89%
of those who report problems go first to a traditional leader

14%
report climate-related problems to the LGA office

19%
attended any climate or environment meeting in the past year

6.6 Willingness, Priorities, and Pledges

Willingness to participate is near-universal and specific. Ninety-nine percent are willing to engage in climate initiatives in the next six months and to attend meetings; 98% would join a committee and attend capacity-building; 97% would contribute labour or time; 94% would contribute small cash or in-kind. Ranked at-risk perceptions place low-income households first (31% of first choices), followed by persons with disabilities (24%), women (22%), the elderly (10%), and youth (8%). Asked to rank ward priorities, respondents put flood control and drainage first (31% of first choices), then early warning systems (21%), awareness workshops (12%), tree planting and land restoration (11%), and water supply/boreholes (10%). Ninety-six percent of respondents made a concrete household pledge: joining monthly clean-ups (36%), planting and maintaining trees (33%), attending governance meetings (18%), or adopting water-saving practices (9%). Town halls (55%) and radio (22%) are the preferred channels for receiving research feedback.



Figure 13. Willingness to participate in climate governance activities (n = 365).

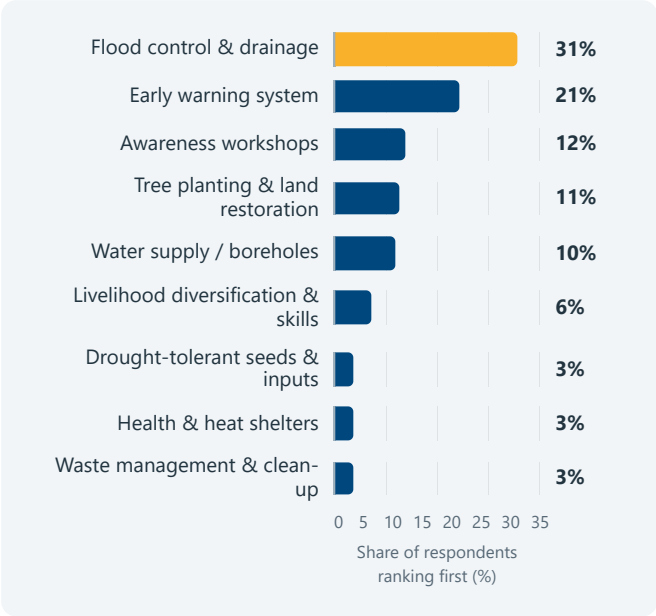


Figure 14. Top-ranked ward priority to reduce climate risk (first choice; n = 333 providing a ranking).

99%
are willing to engage in climate initiatives in the next six months

96%
of respondents made a concrete household pledge

31%
put flood control and drainage first among ward priorities

55%
prefer town halls for receiving research feedback



Photo · Mud-brick homes with peppers drying on the rooftops

SECTION 7

GIS Analysis

7.1 Land Degradation — SDG Indicator 15.3.1

7.2 Land Cover Change (2003–2015 vs 2015–2022)

7.3 Shika Dam Surface Water Dynamics

7.4 Ward-Level Degradation Mapping

GIS ANALYSIS

7. GIS Analysis

7.1 Land Degradation — SDG Indicator 15.3.1

The Trends.Earth assessment computed SDG indicator 15.3.1 for the Sabon Gari analysis extent (5,679 km² assessed; see limitations regarding boundary extent). Between the baseline (2003–2015) and reporting (2015–2022) periods, 39.4% of the assessed land area (2,239 km²) is classified as degraded, 50.2% (2,852 km²) as stable, and 9.1% (519 km²) as improved, with 1.2% lacking data. Degradation is driven almost entirely by the land-productivity sub-indicator: 39.0% of the area shows degraded productivity (moderate decline or declining), while improved productivity covers only 9.2%.

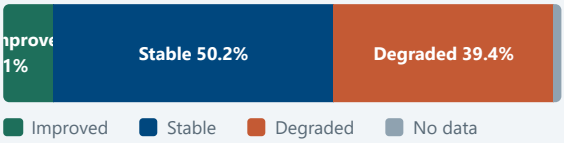
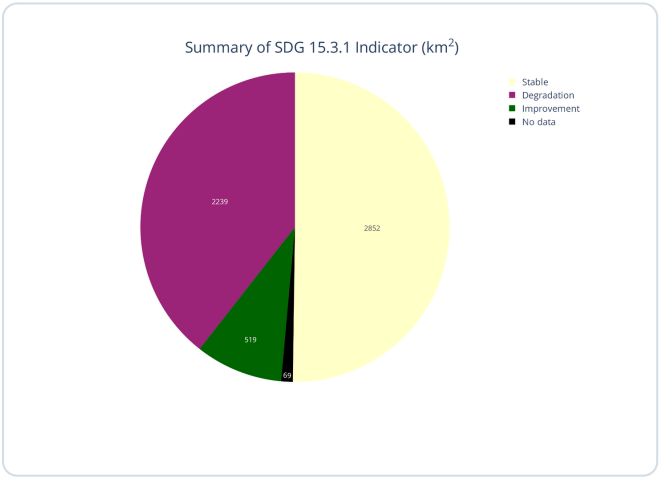
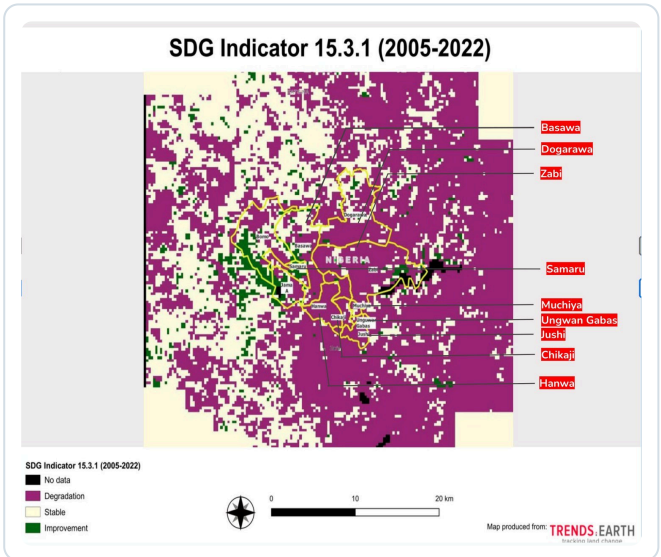


Figure 15. SDG 15.3.1 land condition classes as a share of assessed area (Trends.Earth).

Table 3. *SDG 15.3.1 summary for the Sabon Gari analysis extent (Trends.Earth)*

Land condition	Area (km²)	Share of assessed area
Improved	519	9.1%
Stable	2,852	50.2%
Degraded	2,239	39.4%
No data	69	1.2%
Total assessed	**5,679**	**100%**





Map. SDG Indicator 15.3.1 (2005–2022) for Sabon Gari LGA, with ward locations labelled. Produced with Trends.Earth.

7.2 Land Cover Change (2003–2015 vs 2015–2022)

The land cover analysis reveals rapid urbanisation of cropland. In the baseline period, cropland covered 5,393 km² (94% of the assessed area), artificial surfaces 224 km² (4%), and water bodies 60 km² (1%). By the reporting period, cropland had contracted by 254 km² to 5,139 km² (90%), while artificial surfaces expanded by the same 254 km² to 478 km² (8%) — more than doubling — and water bodies remained unchanged at 60 km². The transition matrices show that the artificial expansion is drawn almost entirely from cropland, and that a large portion of converting land carries declining productivity: approximately 201 km² of cropland-to-artificial transition falls in the "declining productivity" class, with a further 20 km² in moderate decline.

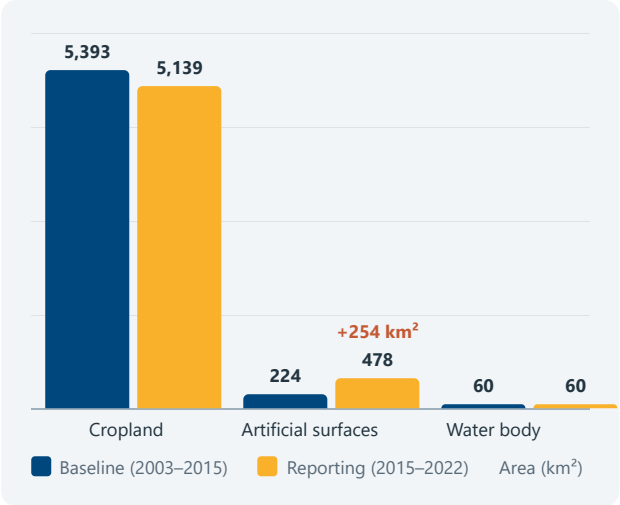
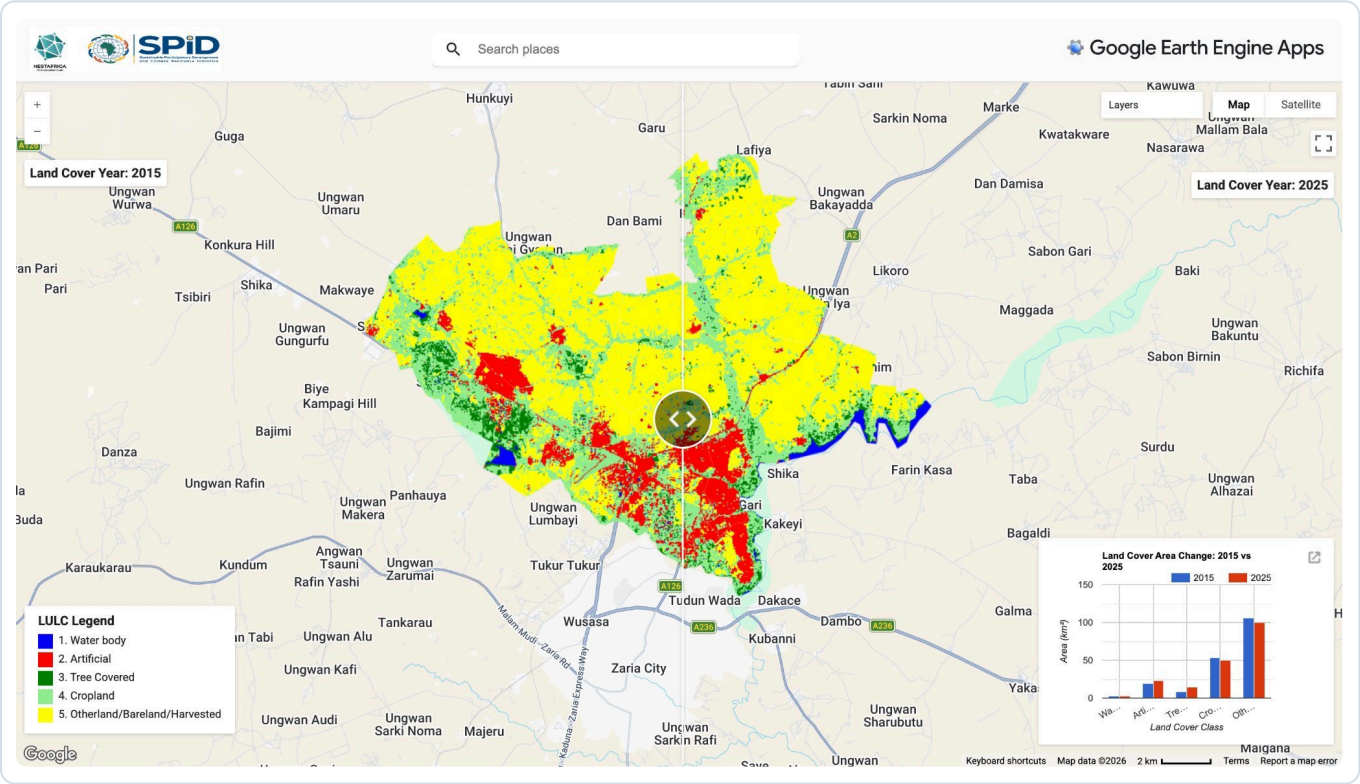


Figure 16. Land cover change between baseline and reporting periods (Trends.Earth).



Map. Land cover in Sabon Gari LGA, 2015 compared with 2025, with area change by land-cover class (Google Earth Engine app).

7.3 Shika Dam Surface Water Dynamics

A satellite time-lapse of the Shika Dam, combining Sentinel-1 SAR and Landsat imagery from 2015 with Google Earth reference imagery of the dam wall, spillways, and downstream reaches, was compiled to examine seasonal water dynamics in the LGA's principal reservoir system. The imagery documents pronounced seasonal variation in inundated area between wet and dry seasons. Because the time-lapse is a visual product without accompanying quantified surface-area statistics in the available dataset, no numerical trend in reservoir extent is claimed; the product corroborates the strongly seasonal hydrology within which the surveyed communities manage water and flood risk, consistent with households' seasonal switch of drinking water sources and the wetland/water-body settings reported around Bomo and Kurmin Bomo.

7.4 Ward-Level Degradation Mapping

The land degradation map differentiates conditions across the mapped ward areas of Basawa, Dogarawa, Zabi, Samaru, Muchiya, Ungwan Gabas, Jushi, Chikaji, and Hanwa. Read alongside the survey, the mapped pattern is consistent with the ward hazard profiles: the denser, more built-up and drainage-adjacent wards where flooding dominates survey responses correspond to areas of artificial-surface expansion, while the cropland-dominant wards where pests, drought, and declining yields dominate correspond to the extensive zones of declining productivity.



Photo. An eroded drainage gully beside a hand-dug well

5,679 km²
assessed in the Trends.Earth analysis extent

+254 km²
expansion of artificial surfaces, drawn from cropland

39.4%
of the assessed land area classified as degraded



Photo · The men's session, seen from another angle

SECTIONS 8–11

Insights

8. Integrated Interpretation of Findings

9. Discussion

10. Key Insights

11. Policy and Practice Implications

INSIGHTS

8. Integrated Interpretation of Findings

Three integrated storylines emerge when the survey and geospatial evidence are read together.

8.1 Urbanising onto floodplains: land conversion and flood exposure

The GIS record shows artificial surfaces more than doubling (+254 km^2) at cropland's expense during 2015–2022, unaccompanied by any expansion of water storage. The survey shows flooding as the leading main hazard (28%), concentrated precisely in the built-up and wetland-adjacent wards (Chikaji, Muchia, Hanwa, Jushi, Angwa Gabas, Jama'a), with 109 respondents describing their ward's land system as "artificial surfaces" and 79 as "wetland." Enumerator-recorded open responses identify blocked drainage and waste accumulation as flood causes. Sealed surfaces increase runoff and drainage congestion; the community's own top-ranked priority—flood control and drainage (31% of first choices) — and its most common adaptation action—waste management and clean-ups—are direct, locally reasoned responses to this dynamic. Survey and satellite point at the same problem from two directions.

8.2 Degrading productivity beneath an agrarian economy

The productivity sub-indicator classifies 39% of the assessed area as declining, on land that remains 90% cropland. The survey population sits directly on this base: 56% of households depend primarily on crop farming, on very small plots (mean 0.96 ha), with 36% reporting below-average yields, intensive input use, and pest/disease outbreaks as the most widely observed hazard (47%), and mean income losses of 36%. Where land productivity declines, small farms with thin buffers absorb the loss as income and food-security stress — visible in the 33% reporting food shortage and 49% reporting crop loss. The degradation map and the household ledger describe the same squeeze.

8.3 A participation vacuum with willing participants

The governance data show a system in which trusted, proximate institutions (traditional leaders, radio, religious networks) carry information and grievances, while formal structures are perceived as unresponsive (33% agree the LGA responds promptly) and opaque (20% agree allocation is transparent), and are barely used (14% report to the LGA office; 20% know a government office to contact). Yet willingness to participate is essentially universal (94-99% across six engagement forms), and 96% made concrete pledges. The constraint is architectural, not motivational: there is no standing ward-level mechanism connecting willing households to formal planning and budgets. This is the operational meaning of the "implementation gap" in Nigeria's participatory provisions at LGA level.

INSIGHTS

9. Discussion

The Sabon Gari evidence aligns closely with the wider literature on climate governance in Nigeria and comparable African contexts. The high-awareness/low-capability pattern (80% aware; 12% high understanding) mirrors findings that awareness campaigns outpace actionable knowledge, and supports the case in community-based adaptation scholarship (e.g., Mercer et al., 2010) for capability-building over messaging. The dominance of radio and traditional leadership as trusted intermediaries is consistent with studies of information ecosystems in Northern Nigeria and with the design logic of Nigeria's NAP Framework, which calls for locally appropriate communication channels. The finding that flood exposure concentrates where artificial surfaces have expanded echoes documented flood-risk dynamics in Kaduna's urbanising corridors, and gives local, data-grounded content to the AU Climate Change and Resilient Development Strategy's emphasis on resilient urbanisation.

The governance results give empirical specificity to analyses of the Climate Change Act (2021), which note weak operationalisation of participation provisions: in Sabon Gari, participation fails not at the level of willingness (99%) or of local voice (65-68% agree leaders listen and women/youth have voice) but at the interface with formal response and finance (33% and 20% agreement). The transparency deficit is particularly consequential for the Act's climate-finance ambitions: communities that do not believe allocations are transparent are unlikely to engage budget processes even when invited. Finally, the gender pattern—women more aware and more active, yet less present in meetings and less educated—cautions against equating attendance with agency and supports gender-intentional design of participation mechanisms rather than open-invitation formats.

Where the data are silent, this report stays silent

Where the data are silent, this report stays silent: without the qualitative component, the study cannot characterise the content of traditional ecological knowledge in Sabon Gari, the internal dynamics of ward decision-making, or the reasons behind the LGA's perceived unresponsiveness. These remain priority questions for the follow-up qualitative work envisaged in the original design.

INSIGHTS

10. Key Insights

01 Climate losses are large and regressive.

Households report losing 36% of income on average to hazards in one year, in an LGA where 54% live on under ₦50,000/month and only 22% hold any financial buffer.

02 Flooding is the flagship risk, and it is partly self-inflicted by land conversion.

Flood-family hazards are the main threat for 41% of respondents, concentrated in wards where the GIS shows artificial surfaces doubling onto former cropland with no drainage or storage response.

03 The land is losing productivity under the LGA's food base.

39% of the assessed area is degraded, while 56% of households depend primarily on crop farming on sub-hectare plots and 36% report falling yields.

04 Awareness is high; capability and institutional connection are low.

80% have heard of climate change, but only 12% claim high understanding; 41% have acted, 19% attended a meeting, and 20% know a government office to contact.

05 Money is the binding constraint on adaptation.

57% of acting households cite lack of money as the main barrier; 100% of non-acting households would adopt practices if supported.

06 Trust runs through traditional leaders and radio, not formal offices.

89% of problem-reporters go first to a traditional leader; 85% of warnings arrive by radio; only 14% report to the LGA and 20% agree allocations are transparent.

07 Women act more but decide less.

Women lead on awareness (85%) and adaptation action (49%) yet attend meetings at barely half the male rate (14% vs 24%) and are twice as likely to lack formal education.

08 The community has already voted with its priorities and pledges.

Flood control/drainage and early warning top the ranked priorities; 96% pledged specific household contributions, led by clean-ups and tree planting.

INSIGHTS

11. Policy and Practice Implications

For the Climate Change Act (2021) and NAP Framework, the findings imply that participation provisions must be operationalised at ward level, through the customary institutions residents already use, rather than through new parallel structures.

For Kaduna State climate and land-use planning, the doubling of artificial surfaces onto cropland without drainage investment implies that development control, drainage master-planning, and wetland protection in the Zaria metropolitan area are now climate-adaptation instruments, not merely urban-planning ones.

For SDG 15.3 (Land Degradation Neutrality) reporting, the Trends.Earth evidence provides a sub-national baseline that state and national reporting can build on once the boundary is verified.

For NiMet and Kaduna SEMA, the 85% radio share of warnings and 43% >7-day lead times indicate that the early warning pipeline works technically; the last-mile failure is advisory content (what to do) and institutional linkage (whom to contact), not signal delivery.

For donors and NGOs, the 100% conditional-adoption finding and the near-universal pledge rate indicate that small, well-targeted adaptation finance (inputs, drainage micro-works, savings-group strengthening) will meet prepared demand rather than needing demand creation.

85%
radio share of early warnings

43%
of warnings arrived with more than seven days of lead time

100%
of non-acting households would adopt practices if supported



Photo · A drainage channel blocked with solid waste

SECTIONS 12–13

Recommendations

12.1 Community Level

12.2 Local Government (Sabon Gari LGA)

12.3 State Government (Kaduna State)

12.4 National Government

12.5 Civil Society

12.6 Development Partners

13. Conclusion

RECOMMENDATIONS

12. Recommendations

12.1 Community Level | 12.2 Local Government (Sabon Gari LGA) | 12.3 State Government (Kaduna State) | 12.4 National Government | 12.5 Civil Society | 12.6 Development Partners

12.1 Community Level

1. The district heads (*Hakimai*) should establish or formalise ward climate committees in all eleven wards, chaired or co-chaired with traditional leadership (*Masu Magri*), with mandated seats for women, youth, and persons with disabilities, converting the 98% committee-willingness into standing structures.
2. The village heads (*Mai Gari*), in collaboration with the ward heads (*Mai Unguwa*), should organise the pledged actions: quarterly ward clean-up rosters (131 households pledged) targeted at drainage lines before and during the rains, and community tree-planting/maintenance campaigns (122 households pledged) linked to woodlot and shade priorities.
3. The village heads/ward heads should encourage and strengthen the development of disaster response teams with the aim of training and mobilization of resources for disaster response.

12.2 Local Government (Sabon Gari LGA)

1. LGA chairman should prioritise flood control and drainage resource allocation in Chikaji, Zabbi, Bomo, and Muchia, the wards combining the highest reported income losses with flood-dominant hazard profiles, including desilting, culvert clearance, and solid-waste management tied to the community clean-up rosters.
2. LGA chairman should liaise with community leaders from the district head's office and make the contact information of the climate/environment desk officer public and easily accessible, publicised by radio and through traditional leaders, to close the 80% gap in knowing whom to contact; route ward committee reports to it on a fixed cycle.
3. LGA chairman should publish simple ward-level climate budget and project information (town-hall postings and radio, matching the 55% town-hall and 22% radio feedback preferences) to address the 57% who perceive allocations as non-transparent.
4. Reinstate agricultural extension presence: only 35% accessed extension advice, in an LGA where pest/disease outbreaks are the most observed hazard and pesticide use is near-universal.

12.1 Community Level | 12.2 Local Government (Sabon Gari LGA) | 12.3 State Government (Kaduna State) | 12.4 National Government | 12.5 Civil Society | 12.6 Development Partners

12.3 State Government (Kaduna State)

1. The Sabon Gari Local Government Council (Planning Department), Kaduna Geographic Information Service (KADGIS), Ministry of Housing and Urban Development, The Kaduna State Environmental Protection Authority (KEPA), and Kaduna State Urban Planning and Development Authority (KASUPDA) should strengthen the enforcement of The Kaduna State Urban and Regional Planning Law (2018) and Kaduna State Geographic Information Service (KADGIS) Law to ensure the Integration of the SDG 15.3.1 evidence into the state's land-use, drainage, and development-control planning for the Sabon Gari metropolitan corridor, explicitly regulating conversion of cropland and wetland to artificial surfaces and requiring drainage provision in approvals. This directly aligns with the Land Use Act of 1978 (Cap L5 LFN 2004) and the Nigerian Urban and Regional Planning Act (CAP N138 LFN 2004).

2. The state government should direct Kaduna State Emergency Management Agency (KADSEMA), Ministry of Agriculture, Kaduna State Environmental Protection Authority (KEPA), and Ministry of Information and Home Affairs to pair every early warning broadcast with actionable household advisories (protect assets, planting-date guidance, hygiene during floods), leveraging the >7-day lead times already achieved as part of the effort to effectively actualise the Kaduna State Climate Change Policy (2024) and the Kaduna State Local Adaptation Plan of Action (LAPA). This directly aligns with the National Strategy for Disaster Risk Reduction (2025 - 2030) and National Disaster Management Framework (NDMF).

3. The Executive Governor of Kaduna State should approved and direct the Ministry of Planning and Budget Commission in collaboration with Ministry of Local Government Affairs & Local Government Councils, Kaduna State Local Government Service Commission, Kaduna State Agricultural Development Agency (KADA) & Rural Water Supply and Sanitation Agency (KADRUWASSA) to allocate fund to a matched small-grants window for ward committee micro-projects (drainage micro-works, boreholes, seedlings), addressing the money barrier that 57% of acting households report, which is in line with with the Kaduna State Local Government Administration Law and the Kaduna State Ward Development Committees (WDCs) Guidelines. This directly aligns with the National Development Plan (NDP) and the Constitution of the Federal Republic of Nigeria (1999, as amended).

4. The Ministry of Human Services and Social Development (MHSSD) should collaborate with the Ministry of Environment and Natural Resources, the Ministry of Local Government Affairs & Ward Development Committees (WDCs), and the Kaduna State Agricultural Development Agency (KADA) to ensure the adoption of gender-intentional participation standards for state-supported climate fora, scheduling, location, and quotas that convert women's higher action rates into decision-making presence. This contributes to the effective actualization of the Kaduna State Climate Change Policy, Kaduna State Gender Policy in Agriculture, Kaduna State Women's Economic Empowerment (WEE) Policy, and Kaduna State Local Adaptation Plan of Action (LAPA), as well as directly aligns with the National Action Plan on Gender and Climate Change (NAPGCC), Revised National Gender Policy, and Nigeria's Nationally Determined Contributions (NDCs).

12.1 Community Level | 12.2 Local Government (Sabon Gari LGA) | 12.3 State Government (Kaduna State) | 12.4 National Government | 12.5 Civil Society | 12.6 Development Partners

12.4 National Government

1. The National Council on Climate Change (NCCC) and Federal Ministry of Environment (Department of Climate Change) should encourage the Climate Change Desk Officers at the State and LGA Level to recognise ward committees and customary institutions as formal consultation channels for adaptation planning and the NAP process. This aligns with the Nigeria Climate Change Act (2021) and National Adaptation Plan (NAP) Framework.
2. Use NiMet's demonstrated radio reach to standardise impact-based, advisory-rich seasonal and event warnings in Hausa for the North-West, and count last-mile advisory uptake, not broadcast alone, as the performance metric.
3. Include verified sub-national SDG 15.3.1 assessments such as this one in Nigeria's Land Degradation Neutrality reporting and target-setting under the UNCCD.

12.5 Civil Society

1. Collaborate with the communities to deliver the capacity-building the community has asked for (98% willing): practical curricula on household risk reduction, climate-smart agriculture, and pest management, delivered through radio programming and ward sessions rather than one-off workshops.
2. Support ward committees with facilitation, simple monitoring tools, and budget-tracking literacy so communities can engage LGA allocations, directly targeting the transparency deficit.
3. Design inclusion deliberately: information in Hausa and accessible formats, meeting times compatible with women's care burdens, and outreach through religious and traditional networks, matching the top participation barriers (information 79%, invitation 47%, time 22%).

12.6 Development Partners

1. Finance a Sabon Gari flood-resilience package combining drainage rehabilitation, waste-management systems, and nature-based measures in the highest-loss wards, co-designed with ward committees.
2. Fund land-restoration and soil-productivity programmes (agroforestry, soil and water conservation, quality inputs) across the degraded cropland zones, building on the 51% already mulching and 33% pledging tree care.
3. Support the completion of the study's qualitative component, focus groups and key informant interviews, including documentation of traditional ecological knowledge, to close the evidence gaps this quantitative report cannot address.
4. Invest in periodic re-measurement (survey and Trends.Earth re-runs on a verified LGA boundary) so that the 2025 dataset becomes a monitored baseline for adaptation outcomes.

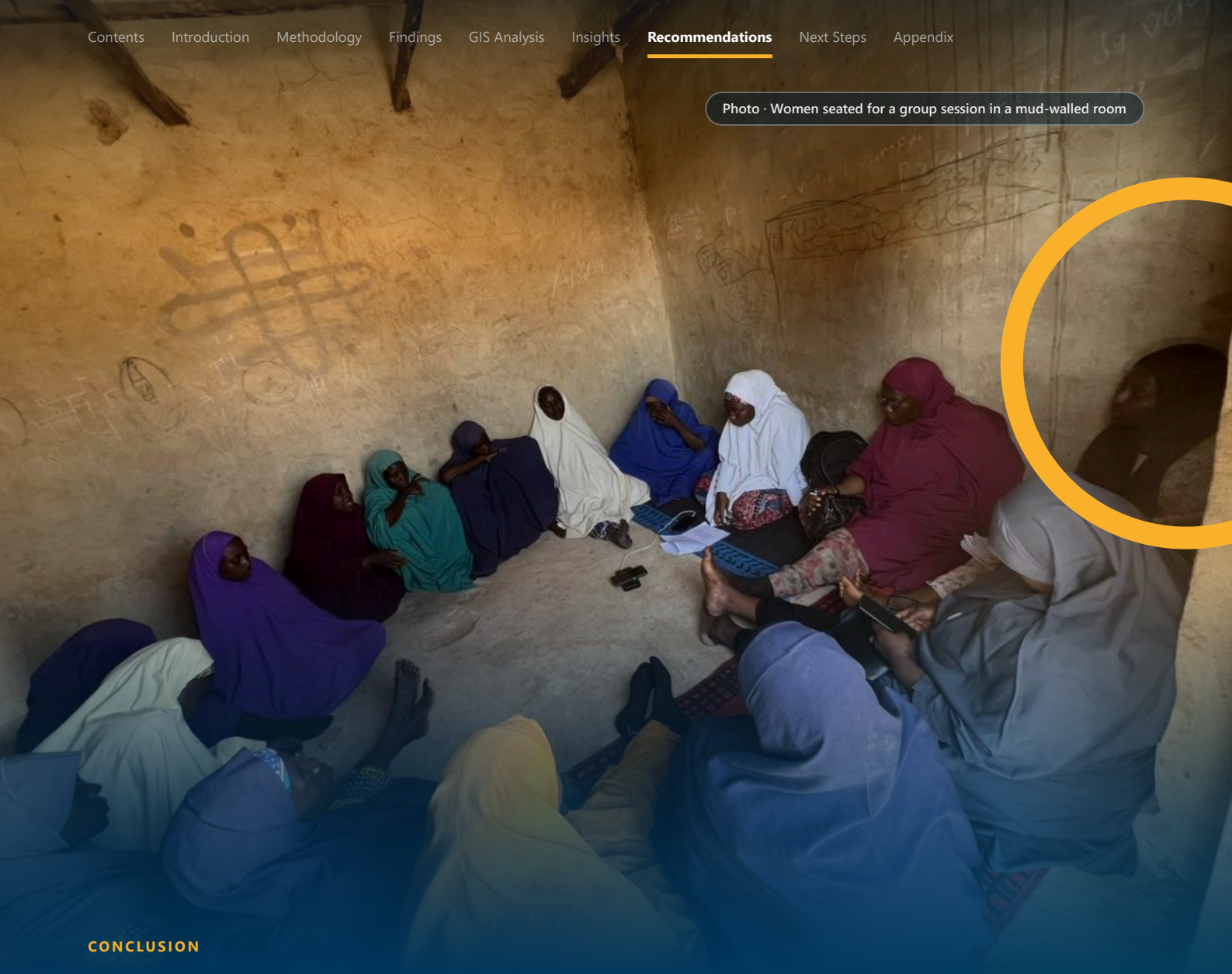


Photo · Women seated for a group session in a mud-walled room

CONCLUSION

13. Conclusion

The CPI4CG assessment set out to give Sabon Gari's residents a place in the evidence base of climate governance, and the data they provided is unambiguous. Climate hazards are already taxing over a third of household income in a population living close to subsistence; the satellite record independently confirms the two physical processes — urbanisation onto drainage land and declining land productivity — that residents experience as floods and failing yields. The community is not waiting passively: awareness is high, adaptation is under way where means allow, priorities are ranked, and pledges are made. What is missing is the connective tissue of governance — ward-level structures, visible institutional contact points, transparent allocation, and modest finance — to convert near-universal willingness into organised, supported action. Building that connective tissue, through the traditional institutions and radio channels residents already trust, is the clearest, most evidence-grounded path from Nigeria's climate policy commitments to changed outcomes in Sabon Gari's eleven wards.

NEXT STEPS

Next Steps and Call for Partnership

The completion of this research marks an important step towards strengthening community participation in climate governance in Sabon Gari Local Government Area. However, generating evidence is only the beginning. For the findings to contribute to meaningful change, they must be shared with the communities that informed the research, translated into actionable recommendations, and brought to the attention of institutions and stakeholders responsible for climate-related decisions and interventions.

Building on the findings of this study, the Sustainable Participatory Development and Climate Resilience Initiative (SPiD), through the Community-led Participatory Initiative for Climate Governance (CPI4CG), seeks to advance the following activities:

Community validation and feedback:

Organise community engagement and validation sessions to present the research findings, obtain feedback from respondents and other community stakeholders, clarify key issues, and ensure that the recommendations reflect local experiences and priorities.

Research dissemination and knowledge sharing:

Develop and disseminate accessible research outputs, including a policy brief, community-friendly findings summary, and visual communication materials, to make the evidence useful to policymakers, community institutions, civil society organisations, researchers, and the wider public.

Multi-stakeholder policy dialogue:

Convene relevant government institutions, traditional and community leaders, civil society organisations, development partners, researchers, youth, women, and other stakeholders to discuss the findings, identify governance gaps, and explore practical ways to strengthen community participation in climate decision-making.

Capacity building and community awareness:

Support community members, particularly women, youth, farmers, and other groups whose livelihoods are affected by climate change, with relevant knowledge and skills on climate change, local adaptation, climate information, environmental sustainability, and participation in climate governance.

Strengthening community–government collaboration:

Facilitate sustained engagement between communities, local authorities, and relevant state institutions to improve access to climate information, strengthen community feedback mechanisms, encourage inclusive planning, and promote the consideration of locally identified priorities in climate-related policies and programmes.

Supporting community-led climate resilience actions:

Translate priority findings into practical, locally appropriate interventions, drawing on community knowledge and indigenous adaptation practices alongside relevant scientific and technical expertise. Potential areas of action will be guided by the research findings and further consultations with affected communities.

Advocacy, institutional engagement, and learning:

Engage relevant institutions to encourage responses to the research recommendations, monitor commitments, document progress, and generate lessons that can inform future climate governance initiatives. The experience from Sabon Gari will also provide an opportunity to explore the adaptation and replication of the CPI4CG approach in other local government areas.



Photo · Men gathered for a community session

NEXT STEPS

Call for Partnership

Achieving these objectives will require collaboration beyond the capacity of any single organisation. SPiD therefore invites government institutions, development partners, civil society organisations, non-governmental organisations, academic and research institutions, climate-focused organisations, foundations, funding agencies, media organisations, and other interested stakeholders to partner in advancing the next phase of CPI4CG.

We welcome partnerships and support in the following areas:

Financial and logistical support:
Resources for community validation meetings, stakeholder dialogues, capacity-building activities, communication materials, and follow-up interventions.

Capacity development:
Technical expertise, facilitators, training resources, and learning opportunities for community members, youth, women, and local institutions.

Policy and advocacy collaboration:
Support for policy engagement, institutional dialogue, evidence-based advocacy, and the promotion of inclusive climate governance.

Community-based implementation:
Collaboration in designing and implementing locally relevant climate resilience and adaptation initiatives.

Knowledge sharing and institutional collaboration:
Opportunities to exchange expertise, share lessons, strengthen partnerships, and explore the replication of the CPI4CG approach in other communities.

SPiD is particularly interested in building partnerships that go beyond one-off engagements and contribute to sustained community participation, institutional responsiveness, and locally led climate action.

> **Our goal is to ensure that the experiences, knowledge, and priorities of communities are not only documented but also recognised and reflected in the decisions, policies, and programmes that affect their lives.**

We invite interested organisations and institutions to join SPiD in moving from community-generated evidence to inclusive policy engagement and practical climate resilience action. Together, we can help ensure that climate governance is informed by the people most affected by climate change and responsive to the realities of their communities.

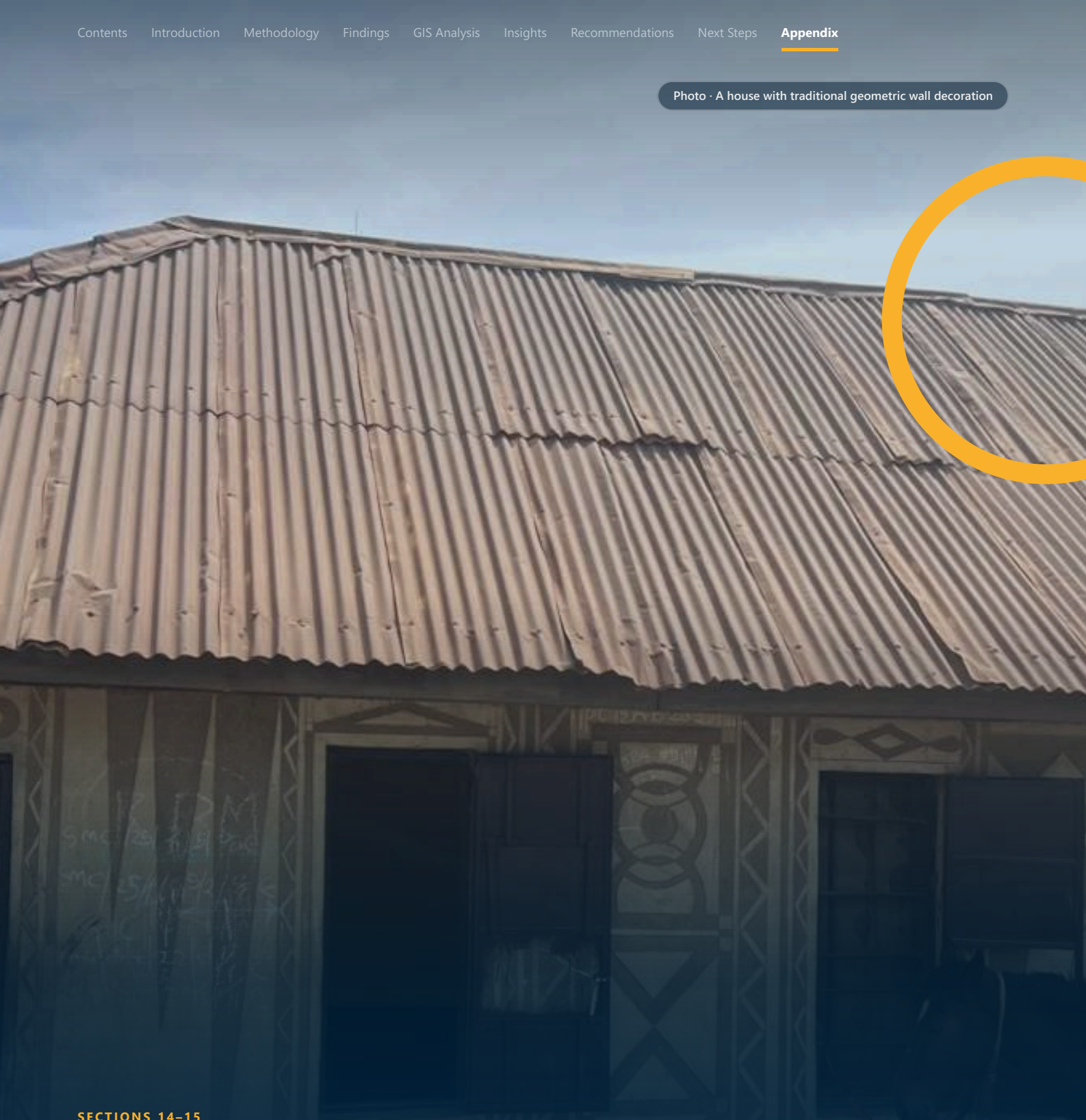


Photo · A house with traditional geometric wall decoration

SECTIONS 14–15

Appendix

14. References

15. Annexes

Annex A. Survey Modules Covered

Annex B. Governance Perception Detail (n = 365)

Annex C. Data and Reproducibility

APPENDIX

14. References

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15. Annexes

Annex A. Survey Modules Covered

Consent and recontact; interview metadata and geopoint; respondent and household demographics; livelihoods and income; disability and female headship; land access; drinking water by season and water-fetching times; sanitation, cooking energy, electricity, assets; climate awareness and self-rated understanding; knowledge of government offices; early warning access, sources, lead time, and preferred channels; ward land-use system; hazards observed and main hazard, degree, drivers, impacts, income loss, and perceived frequency; adaptation actions, barriers, effectiveness, and conditional adoption; local groups, meeting attendance and effectiveness; five governance perception statements; reporting pathways; six willingness items; agriculture (farm size, irrigation, yields, inputs, soil conservation, extension, land/water conflict); water insecurity days; illness and heat effects; insurance and savings; ranked at-risk groups and ward priorities; household pledges; and preferred feedback channels.

Annex B. Governance Perception Detail (n = 365)

Table 4. *Full response distribution on governance statements*

Statement ↕	Agree/Strongly agree ↕	Neutral/Don't know ↕	Disagree/Strongly disagree ↕
Local leaders listen to community concerns	64.7%	17.5%	17.8%
Women and youth have a voice in decisions	68.5%	18.6%	12.9%
Information from local authorities is reliable	64.7%	18.1%	17.3%
LGA responds promptly to climate impacts	33.2%	20.0%	46.8%
Climate resource allocation is transparent	20.0%	22.7%	57.3%

Annex C. Data and Reproducibility

Survey microdata: spid_analytics_dashboard.xlsx (365 records, 103 variables, 13 multiple-response tables). GIS: SDG15_3_1_summary_sabon_gari_.xlsx (Trends.Earth SDG 15.3.1, productivity, land cover, and soil carbon tables) and the SPiD/Nest Africa GIS assessment report. All percentages in this report are reproducible from these files; bases are stated where they differ from n = 365.

SPiD
Sustainable Participatory Development and Climate Resilience Initiative

NESTAFRICA
AI Innovation Lab

Community-led Participatory Initiative for Climate Governance (CPI4CG)

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With technical support on GIS, Data Management, and Analysis from Nest Africa Artificial Intelligence Innovation Lab

Field data: November–December 2025 • Report: July 2026

